UDROŻNIENIE ŁÓDZKIEGO WĘZŁA KOLEJOWEGO (TEN-T), ETAP II, ODCINEK ŁÓDŹ FABRYCZNA - ŁÓDŹ KALISKA/ŁÓDŹ ŻABIENIEC – PROJEKT POIIŚ 5.1-15

TBM 13.08 M - CONSTRUCTION FOLLOW-UP

IDENTIFICATION TABLE	
Client	PBDiM - Energopol
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1. INTRODUCTION

S-1222 TBM EPB machine with 13.08 m excavation diameter is currently mining Axis 23, in the section between Polesie Station and Śródmieście Station. TBM has restarted from chainage 1+962.

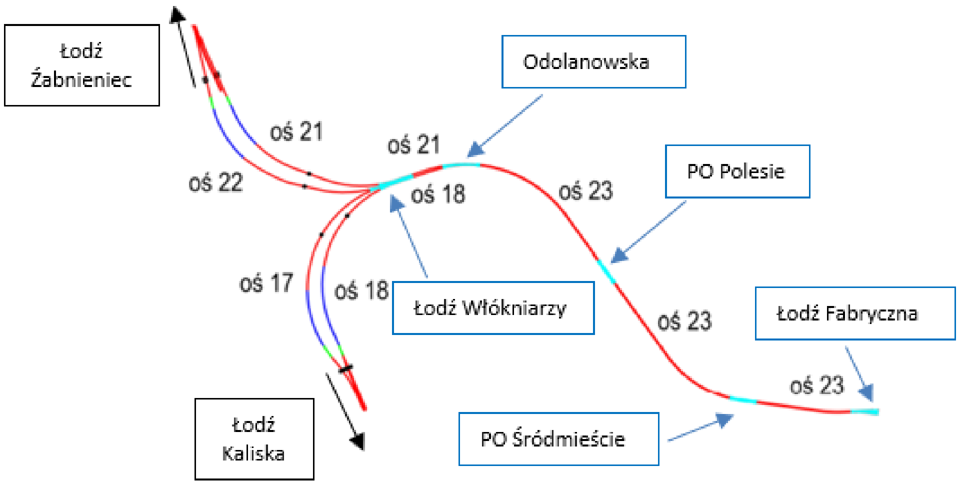

Figure 1-1 Overview of main structural elements of the Project

This daily follow-up report aims to give a general overview of S-1222 TBM drive during 02/09/2024. It includes EPB face pressure, belt scale parameters, thrust and torque values, primary grout parameters, guidance, as well as monitoring follow-up data.

According to VMT system, TBM is at Chainage 1+647.54 after the erection of ring n. 257; which corresponds to a total excavation length of 318.93 m from the starting chainage.

Figure 1-2 Plan view of TBM Position on 03/09/2024

2. FACE PRESSURES AND EXCAVATION VOLUMES

Average EPB pressure at the crown is at about 1.7 bar. The value is within the attention values.

There are drops of face pressure during the excavation of ring 253, 254, 255, 256, with values that range between 1.3 bar and 1.4 bar, lower than alarm thresholds. The pressure has then recovered rapidly within attention limits.

Figure 2-1 shows EPB sensors.

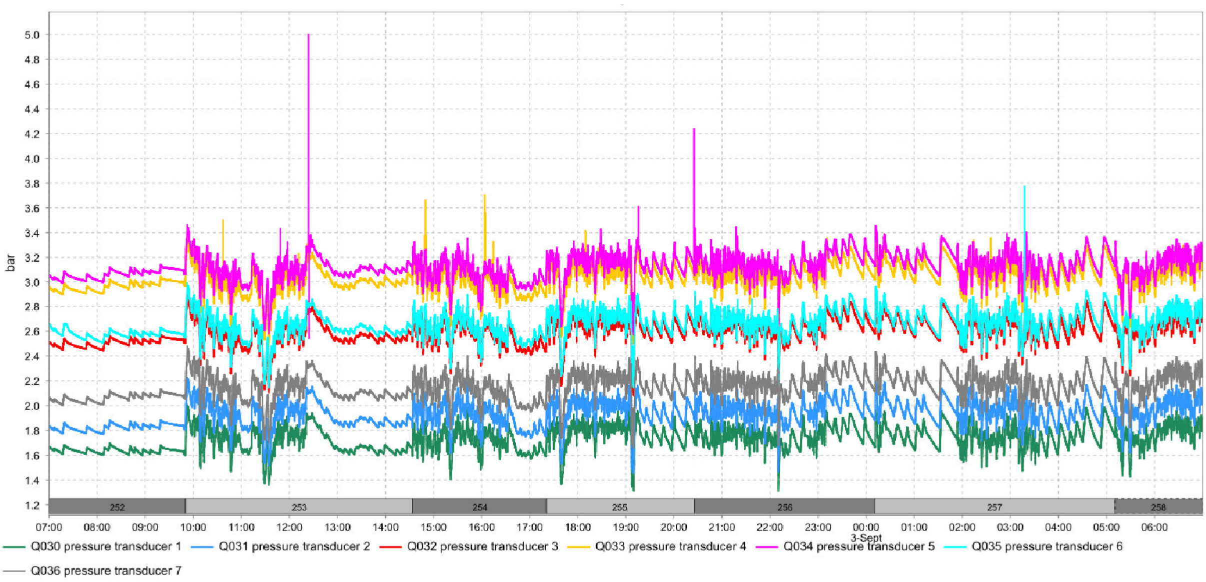

Figure 2-1 EPB pressure sensor

The delta between pressure transducer 1 and 2 has an average value of 0.2 bar indicating that the bulk chamber is correctly filled with material.

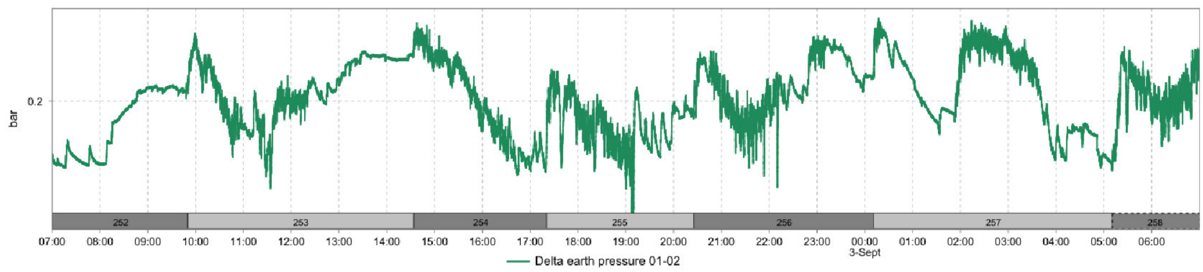

Figure 2-2 Delta between pressure sensors 1 and 2

Belt scale n. 2 show values lower than n. 1 but similar to each other. The belt scale value is a raw indication of the excavated material quantity.

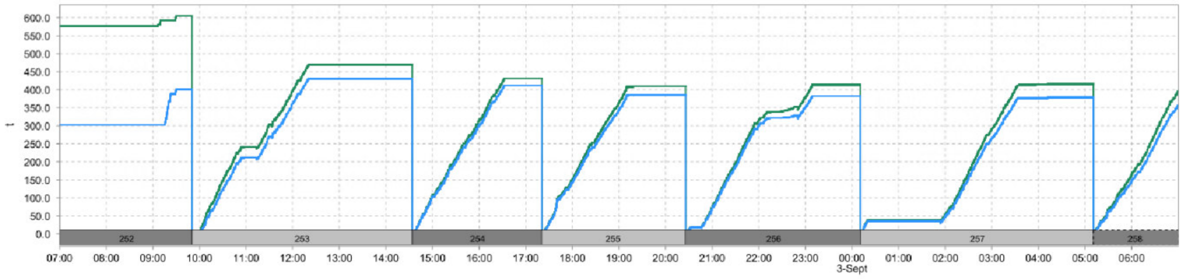

Figure 2-3 Belt scales

Table 2-1 Belt scale values at the end of each advance

Ring	Belt scale 1 (t)	Belt scale 2 (t)
252	605	401
253	469	431
254	431	412
255	410	386
256	413	383
257	416	378

3. THRUST AND CUTTERHEAD TORQUE

During excavation, the average torque is of 9 MNm.

During advancement, average advance rate is of 12 mm/min.

During excavation, thrust force shows an average value of 57 MN and a maximum of 76.5 MN.

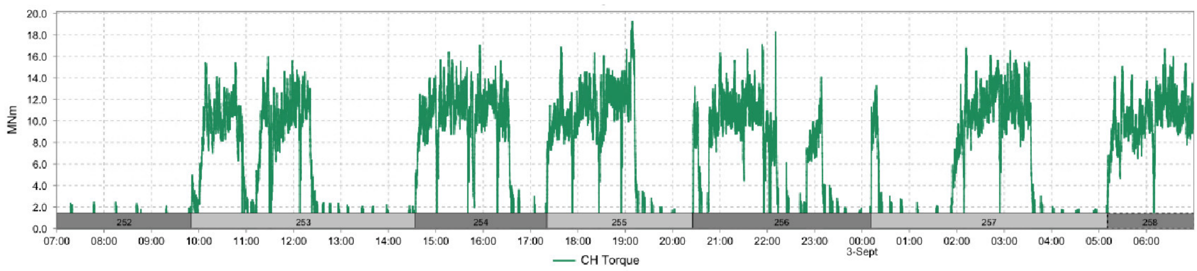

Figure 3-1 CH Torque

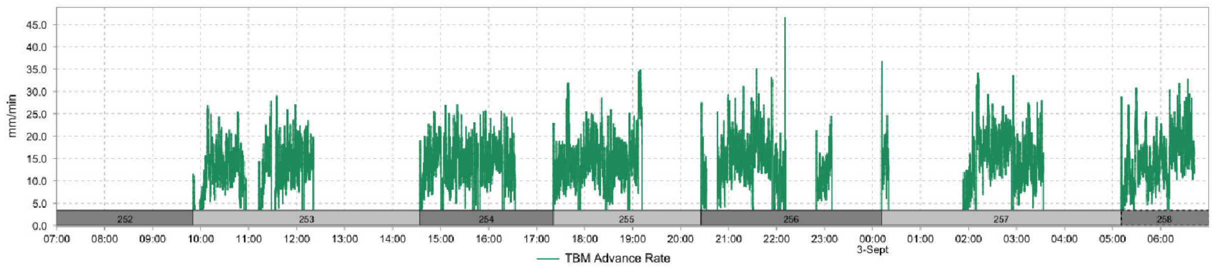

Figure 3-2 Advance rate

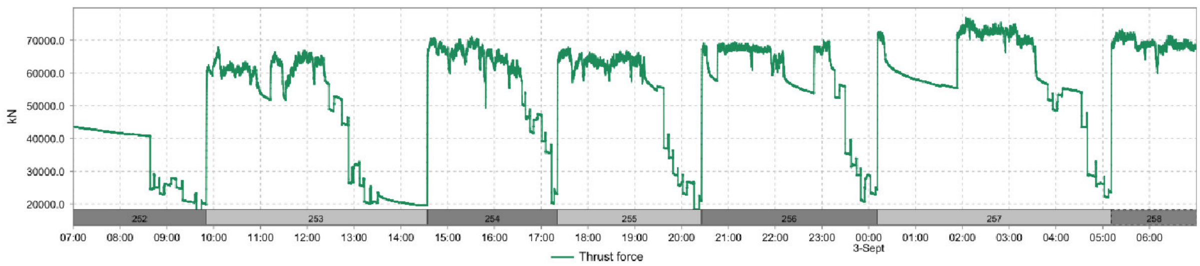

Figure 3-3 Thrust force

4. GROUT INJECTION

Grout injection has a volume below the minimum theoretical annular volume +%10 of 13.54 m^3. Ratio B/A is at 7%.

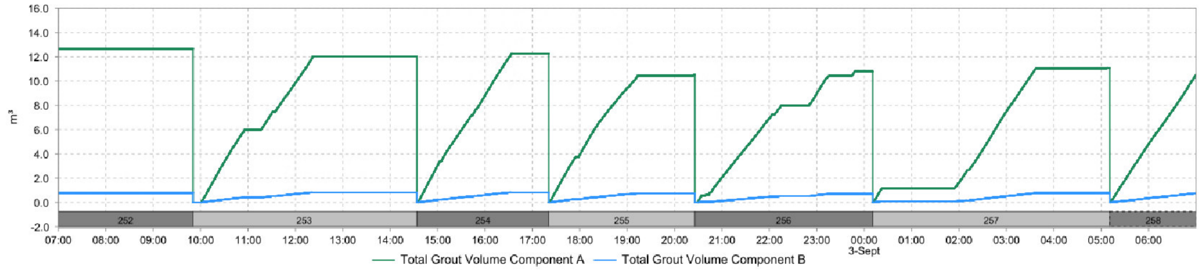

Figure 4-1 Grout volumes

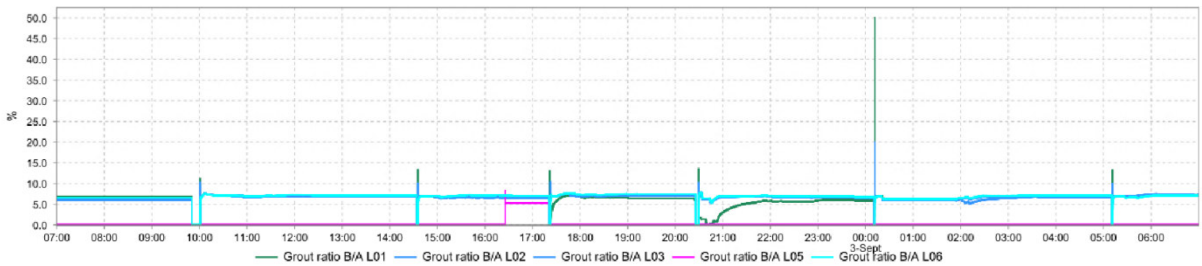

Figure 4-2 Volume ratio Components B/A

Table 4-1 Grout volumes per ring

Ring	Component A (m^3)	Component B (m^3)	B/A (%)
252	12.7	0.8	6
253	12	0.8	7
254	12.3	0.8	7
255	10.5	0.7	7
256	10.8	0.7	6
257	11.1	0.8	7

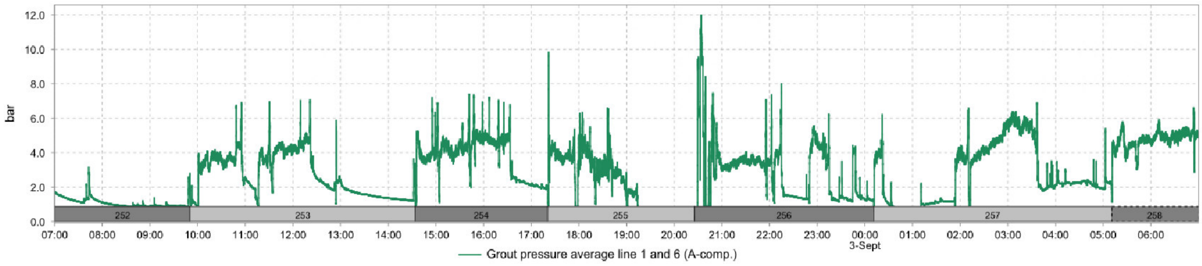

Figure 4-3 Average injection pressure line 1 and 6

5. GUIDANCE

The TBM vertical deviation is of about 0.68 mm on the front edge of the shield and the horizontal deviation of about -4.26 mm (Figure 5-1). Maximum radial deviation is of 29.91 mm.

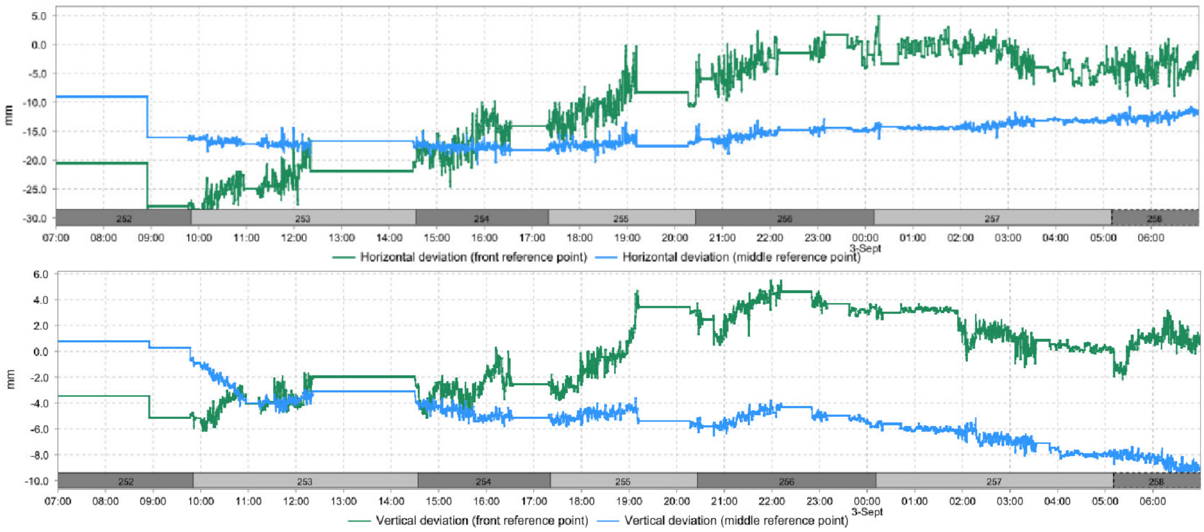

Figure 5-1 Machine deviation from the alignment

6. MONITORING

According to Geoscope, settlements of the buildings above the tunnel are kept within the limits.

Figure 6-1 Monitoring points

The monitoring points on Prochnika street are stable.

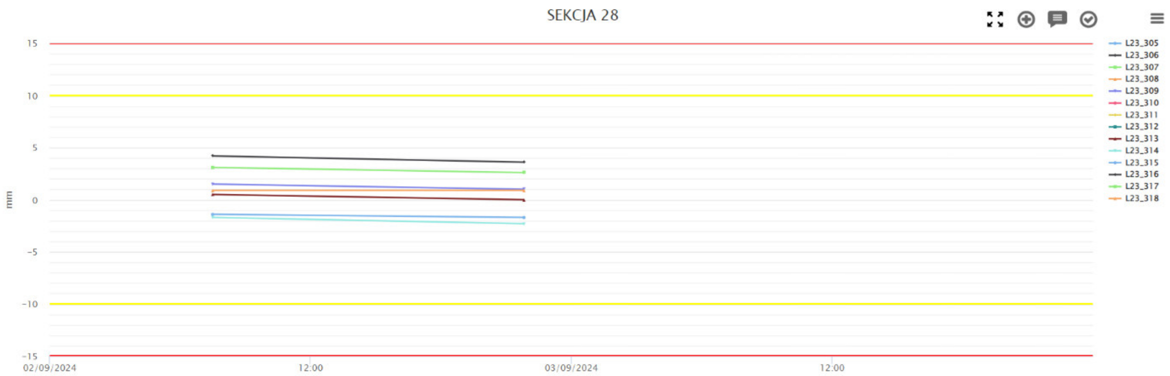

Figure 6-2 Settlement of Section on Prochnika street

On PR0440, monitoring points show that the downward trend previously observed tends to stabilize. An uplift can be observed for point PR0440-05R.

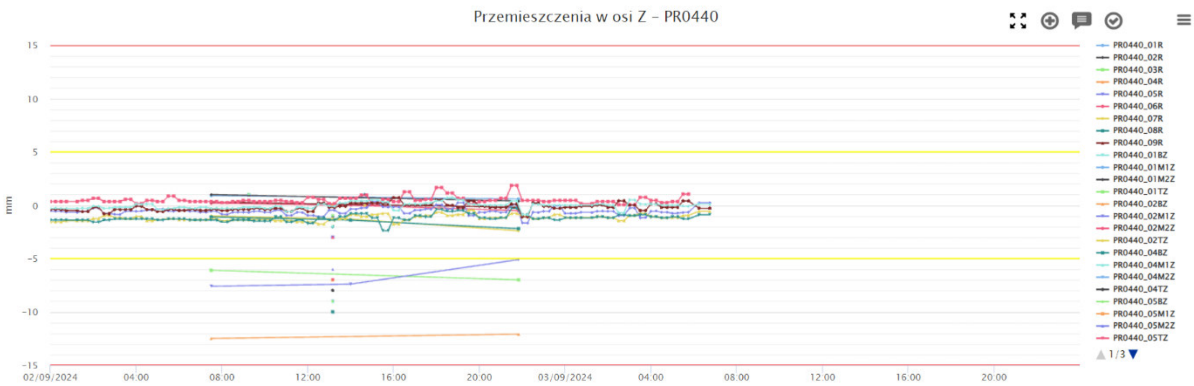

Figure 6-3 Settlement of building PR0440

On PR0460, monitoring points are stable.

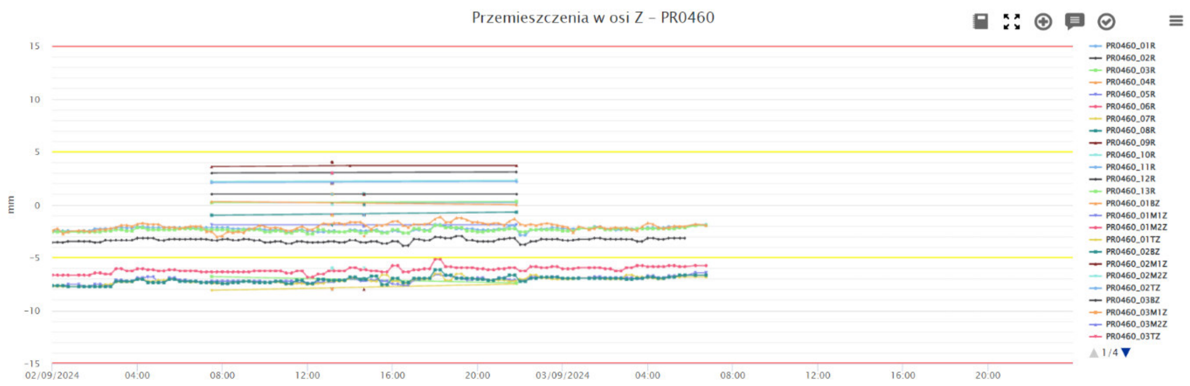

Figure 6-4 Settlement of building PR0460

On MA0210, monitoring points are stable.

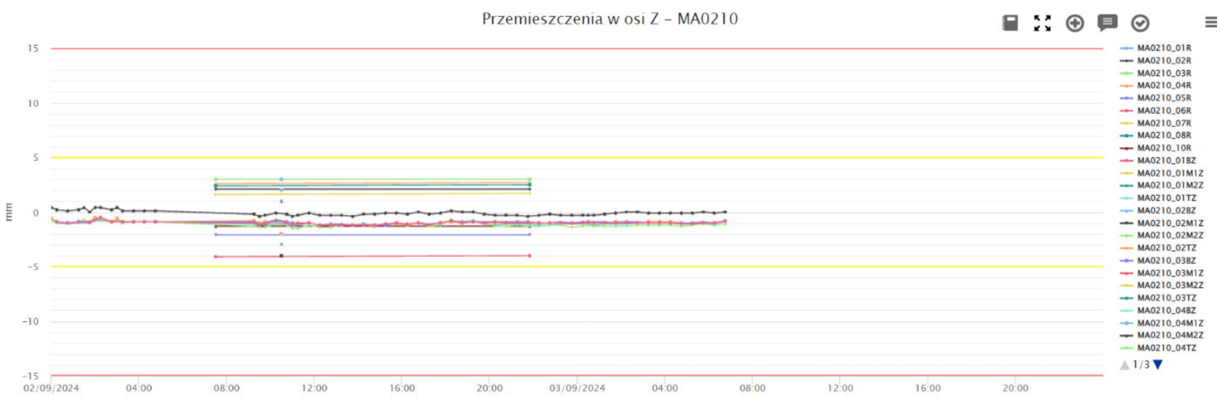

Figure 6-5 Settlement of building MA0210

On MA0230, monitoring points are stable.

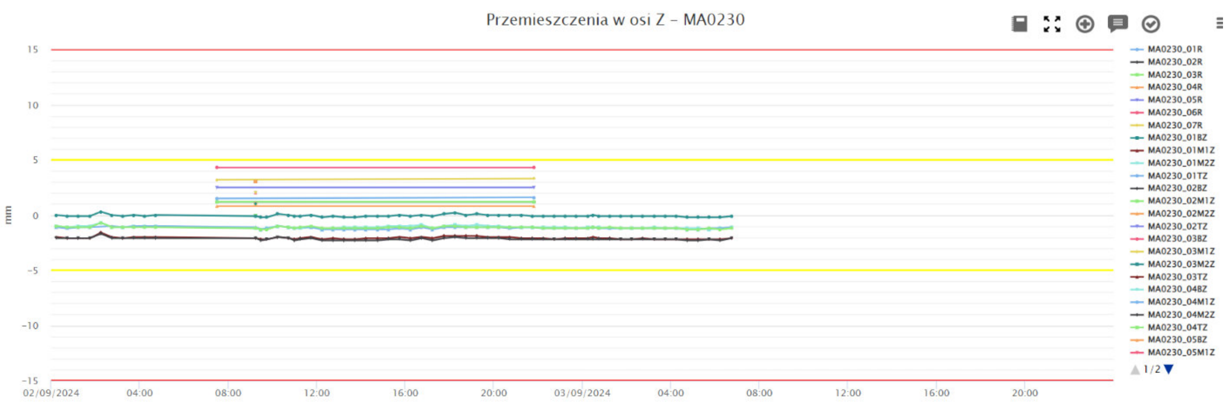

Figure 6-6 Settlement of building MA0230

7. COMPENSATION GROUTING ACTIVITIES

According to Raport Dobowy 02/09/2024 of Soletanche, compensation grouting has been performed on the 02/09/2024 from Shaft 3 in Próchnika 42.

The total amount of declared injected volume is 36953.67 l.

8. SOIL CONDITIONING

The volume of injected foam is plotted in the following Figure. The FIR value is computed based on the excavated material amount per ring and reported in the Table.

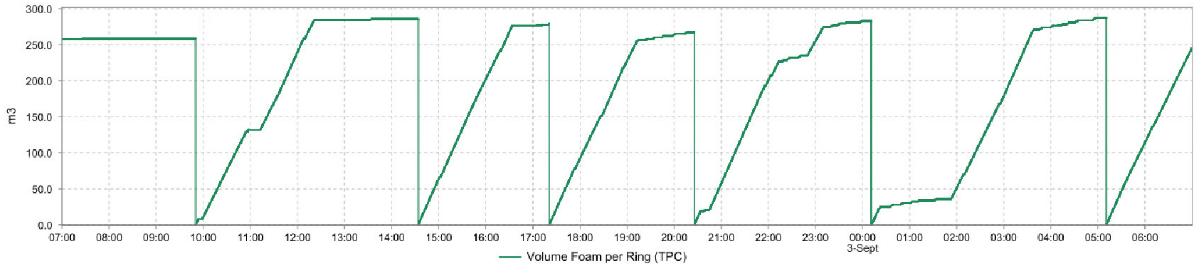

Figure 8-1 Injected foam volume

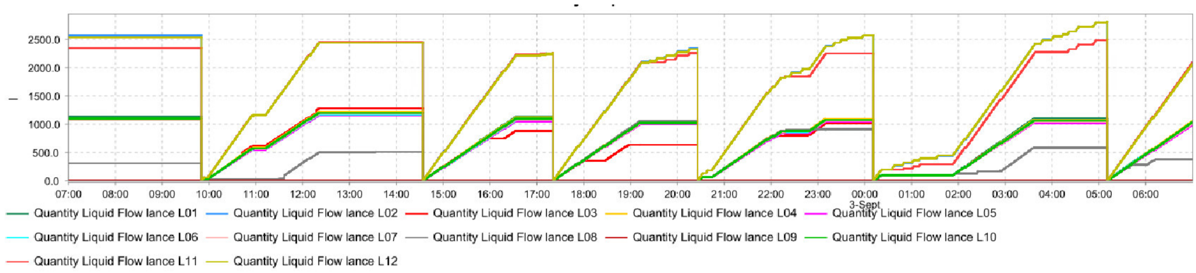

Figure 8-2 Conditioning liquid volume

Table 8-1 Soil conditioning data

Ring	Liquid volume (m^3)	C (%)	FIR (%)
252	15.5	2	123
253	16.2	2	136
254	15.4	2	132
255	14.8	2	127
256	0	0	127
257	16.1	2	137